

Lab 3: RLC Circuits - Part 1

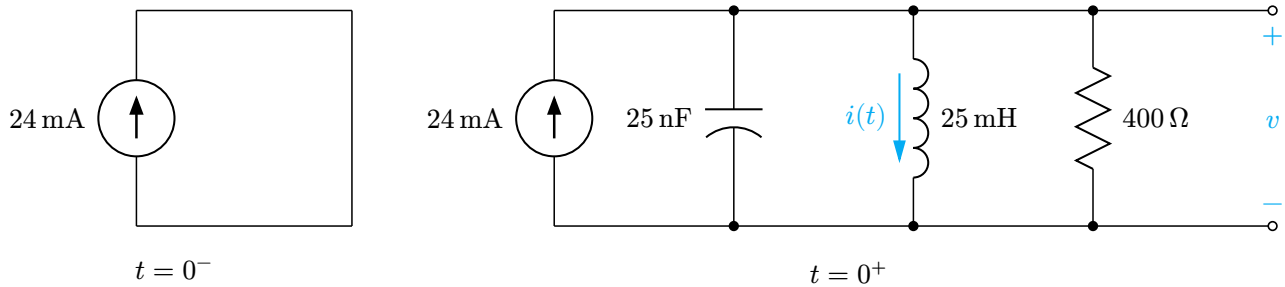
ECE2200L Section 04, Fall 2026

Matthew Ponciano, Eli Pantoja

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No 1.

Step 1. There is no initial energy stored in this circuit. find $i(t)$ for $t \geq 0$.



$$v(0^+) = v_C(0^+) = v_C(0^-) = 0 \text{ V}$$

$$i_s = i_C + i_L + i_R$$

$$i_s = C \frac{dv}{dt} + i_L + \frac{v}{R}$$

$$0 = C \frac{d^2v}{dt^2} + \frac{di_L}{dt} + \frac{1}{R} \frac{dv}{dt}$$

$$0 = C \frac{d^2v}{dt^2} + \frac{1}{L}v + \frac{1}{R} \frac{dv}{dt}$$

$$0 = \frac{d^2v}{dt^2} + \frac{1}{RC} \frac{dv}{dt} + \frac{1}{LC}v$$

$$0 = v'' + \frac{1}{RC}v' + \frac{1}{LC}v$$

recall for a parallel RLC system:

$$v'' + 2\alpha v' + \omega_o^2 v = 0$$

so,

$$\alpha = \frac{1}{2RC} \quad \text{and} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\alpha = 50000 \text{ s}^{-1} \quad \text{and} \quad \omega_0 = 40000 \text{ rad/s}$$

and since $\alpha > \omega_0$, the system is overdamped, which implies

$$s \in -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$s \in \{-20000, -80000\}$$

recall the solution is of the form

$$v(t) = Ae^{s_1 t} + Be^{s_2 t}$$

so

$$v(t) = Ae^{-20000t} + Be^{-80000t} \text{ V}$$

to find A and B , use the initial condition $v(0^+) = 0$, which implies

$$A + B = 0$$

and the KCL equation at $t = 0^+$. since both $v(0^+)$ and $i_L(0^+)$ are 0,

$$\begin{aligned} 24 \text{ mA} &= (25 \text{ nF}) \frac{dv(0^+)}{dt} \\ \frac{dv(0^+)}{dt} &= \frac{24 \text{ mA}}{25 \text{ nF}} \\ \frac{dv(0^+)}{dt} &= 960000 \text{ V/s.} \end{aligned}$$

differentiating the solution,

$$v'(t) = -20000Ae^{-20000t} - 80000Be^{-80000t}$$

and at $t = 0$,

$$-20000A - 80000B = 960000.$$

solving the system yields

$$\begin{aligned} \begin{bmatrix} 1 & 1 \\ -20000 & -80000 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} &= \begin{bmatrix} 0 \\ 960000 \end{bmatrix} \\ \begin{bmatrix} A \\ B \end{bmatrix} &= \begin{bmatrix} 16 \\ -16 \end{bmatrix}. \end{aligned}$$

thus,

$$v(t) = 16e^{-20000t} - 16e^{-80000t} \text{ V for } t \geq 0$$

now recall

$$i_L(t) = \frac{1}{L} \int_0^t v(\tau) d\tau + i_L(0^+)$$

integrating

$$i_L(t) = \frac{16}{L} \int_0^t e^{-20000\tau} - e^{-80000\tau} d\tau + 0$$

we obtain

$$\boxed{i_L(t) = 24 - 32e^{-20000t} + 8e^{-80000t} \text{ mA}} \text{ for } t \geq 0$$

Step 2. Simulate the circuit to confirm your result in Step 1.

[View simulation code](#)

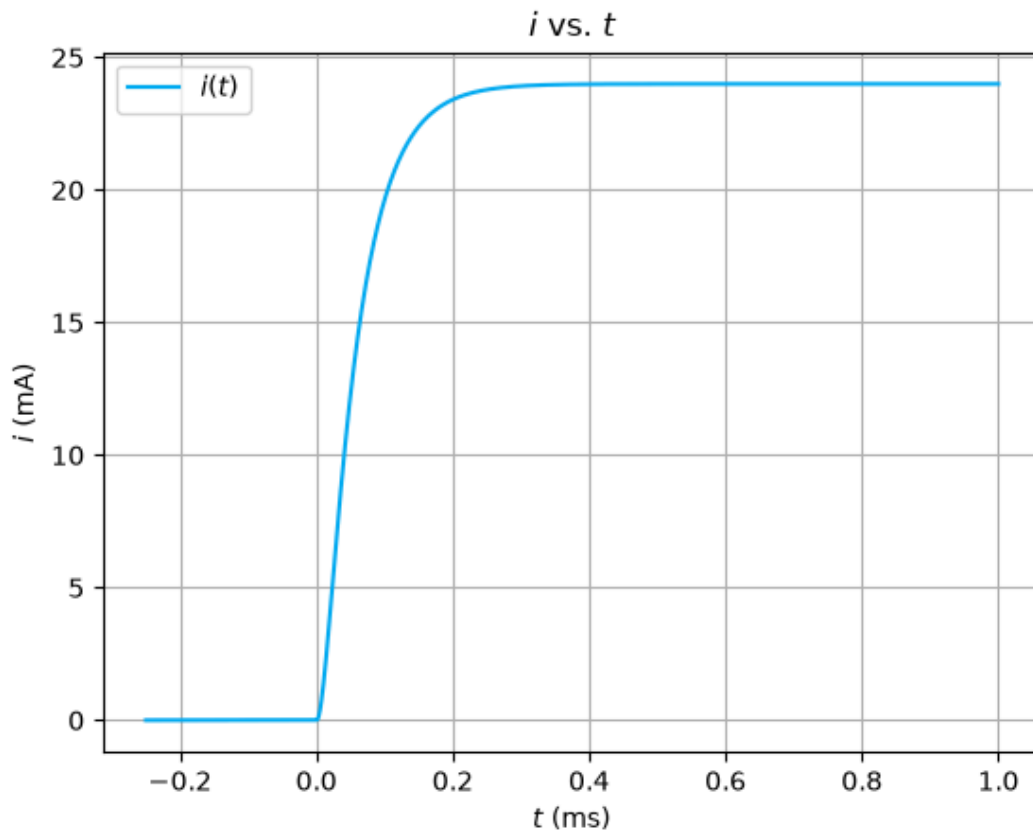
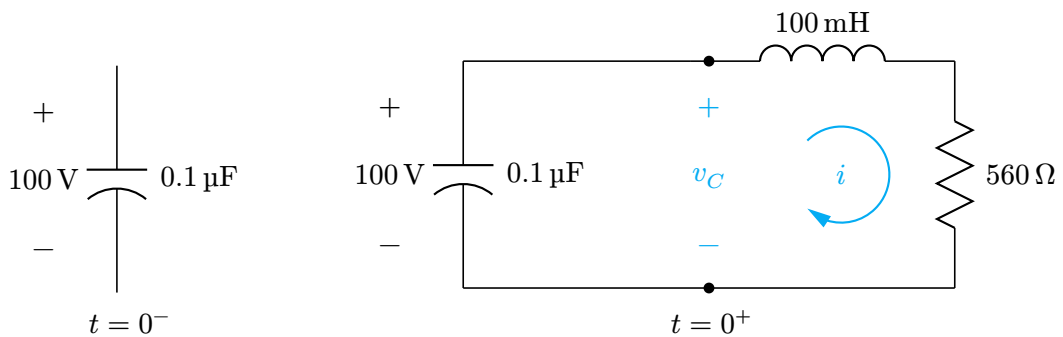


Figure 1: Simulated inductor current $i(t)$, confirming the overdamped response predicted analytically. The current begins at 0mA and approaches the steady-state value of 24mA without oscillation.

No 2.

Step 1. The capacitor is charged to 100 V and at $t = 0$, the switch closes. Find $i(t)$ for $t \geq 0$.



$$i(0^+) = i_L(0^+) = i_L(0^-) = 0 \text{ A}$$

$$\Rightarrow v_C(0^+) = 100 \text{ V}$$

$$v_C = v_L + v_R$$

$$v_C = L \frac{di}{dt} + Ri$$

$$\frac{dv_C}{dt} = L \frac{d^2i}{dt^2} + R \frac{di}{dt}$$

since i flows out the capacitor's positive terminal, $i = -C \frac{dv_C}{dt}$

$$-\frac{1}{C}i = L \frac{d^2i}{dt^2} + \frac{di}{dt}R$$

$$0 = \frac{d^2i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC}i$$

$$0 = i'' + \frac{R}{L}i' + \frac{1}{LC}i$$

recall for a series RLC system:

$$i'' + 2\alpha i' + \omega_0^2 i = 0$$

so,

$$\alpha = \frac{R}{2L} \quad \text{and} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\alpha = 2800 \text{ s}^{-1} \quad \text{and} \quad \omega_0 = 10000 \text{ rad/s}$$

and since $\alpha < \omega_0$, the system is underdamped, which implies the solution is of shape

$$i(t) = e^{-\alpha t} [A \cos(\omega_d t) + B \sin(\omega_d t)]$$

where

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = 9600$$

so

$$i(t) = e^{-2800t} [A \cos(9600t) + B \sin(9600t)]$$

now solve for A and B by setting $t = 0$. since $i(0^+) = i(0^-)$,

$$A = 0$$

looking at the KVL equation at $t = 0^+$

$$100 \text{ V} = (100 \text{ mH}) \frac{di(0^+)}{dt}$$

$$\frac{di(0^+)}{dt} = 1000 \text{ A/s}$$

differentiating the solution and evaluating at 0 yields

$$i'(0) = 9600B$$

$$B = \frac{1000}{9600} \approx 104.167 \text{ mA}$$

thus,

$$i(t) = \frac{1000}{9600} e^{-2800t} \sin(9600t) \text{ A for } t \geq 0$$

Step 2. Simulate the circuit to confirm your result in Step 1.

[View simulation code](#)

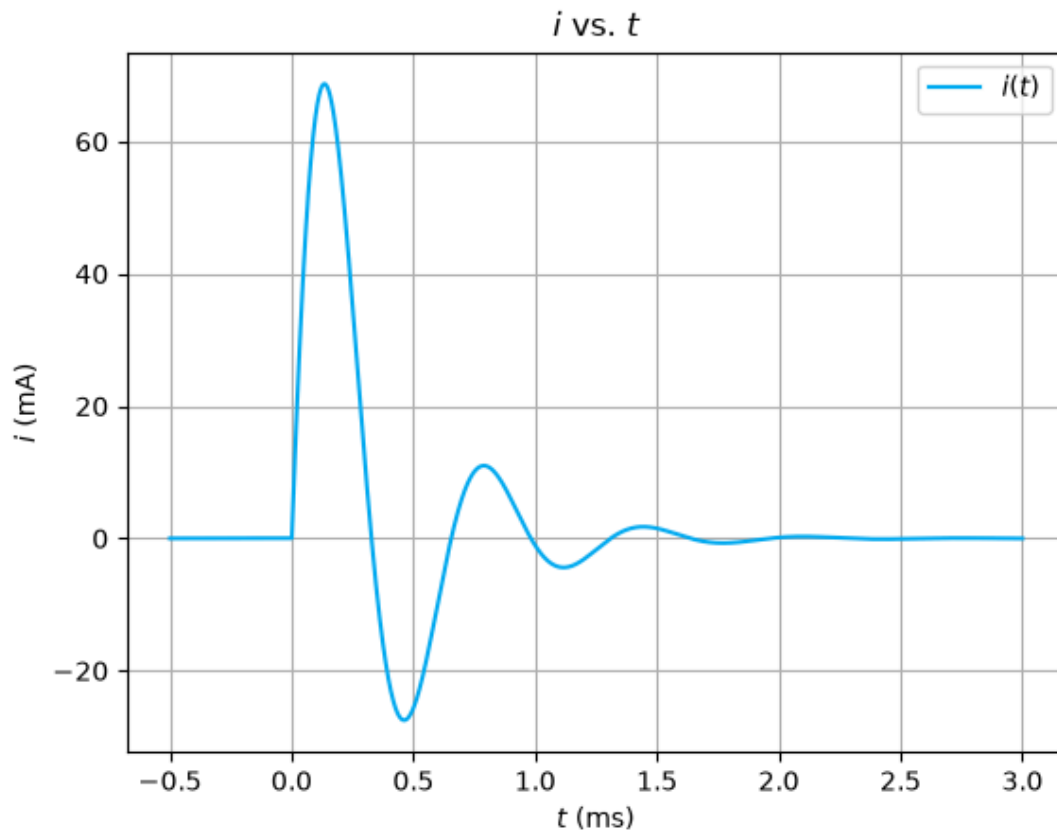
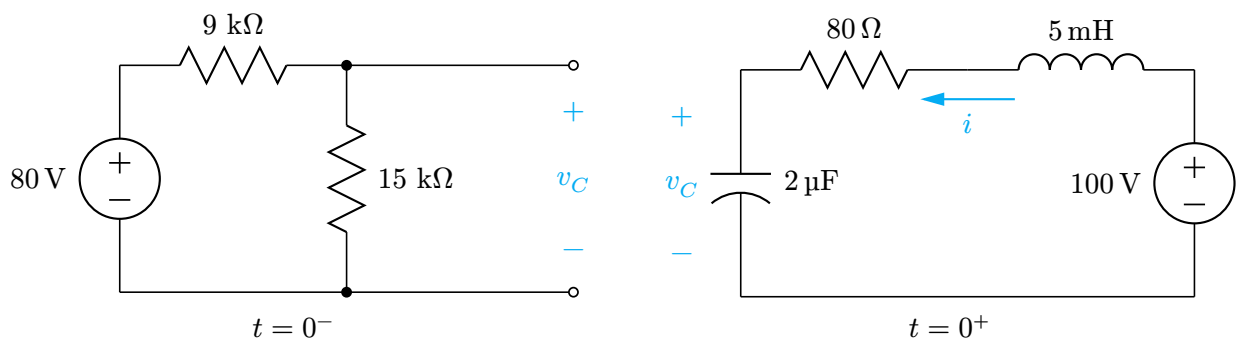


Figure 2: Simulated loop current $i(t)$, showing the expected underdamped natural response of the series RLC circuit. The current starts at zero, oscillates with exponentially decreasing amplitude, and eventually decays back to 0 A as the initially stored capacitor energy is dissipated by the resistor.

No 3.

Step 1. Find $v_C(t)$ for $t \geq 0$



find $v_C(0^-)$ using voltage division:

$$v_C(0^-) = (80 \text{ V}) \frac{15 \text{ k}\Omega}{15 \text{ k}\Omega + 9 \text{ k}\Omega} = 50 \text{ V}$$

then, by continuity,

$$v_C(0^+) = v_C(0^-) = 50 \text{ V}$$

the inductor branch was open before switching, so

$$i(0^-) = 0 \text{ A}$$

and therefore,

$$i(0^+) = i(0^-) = 0 \text{ A.}$$

now construct the differential equation:

$$v_L + v_R + v_C = 100$$

$$L \frac{di}{dt} + Ri + v_C = 100$$

$$L \frac{d^2i}{dt^2} + R \frac{di}{dt} + \frac{dv_C}{dt} = 0$$

$$L \frac{d^2i}{dt^2} + R \frac{di}{dt} + \frac{1}{C}i = 0$$

$$i'' + \frac{R}{L}i' + \frac{1}{LC}i = 0$$

$$i'' + \frac{R}{L}i' + \frac{1}{LC}i = 0 \iff i'' + 2\alpha i' + \omega_0^2 i = 0$$

$$\Rightarrow \alpha = \frac{R}{2L} = 8000 \text{ s}^{-1}$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}} = 10000 \text{ rad/s}$$

since $\alpha < \omega_0$, the system is underdamped, which implies the solution is of shape

$$i(t) \sim e^{-\alpha t} [A \cos(\omega_d t) + B \sin(\omega_d t)], \quad \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

recall $i(0^+) = 0 \text{ A}$; substituting into $i(t)$ yields

$$A = 0$$

now find $i'(t)$ at $t = 0^+$:

$$L \frac{di(0^+)}{dt} + \cancel{iR}^0 + \cancel{v_C}^{50} = 100$$

$$i'(0^+) = \frac{50}{5 \text{ mH}}$$

$$i'(0^+) = 10000 \text{ A/s}$$

differentiating and evaluating $i(t)$ at $t = 0$ yields

$$i'(0^+) = -\alpha A + \omega_d B$$

and since $A = 0$,

$$B = \frac{i'(0^+)}{\omega_d}$$

$$B = \frac{5}{3} \approx 1.667 \text{ A}$$

so the current around the RLC loop is

$$i(t) = \frac{5}{3} e^{-8000t} \sin(6000t) \text{ A for } t \geq 0$$

to find $v_C(t)$, recall that

$$v_C(t) = \frac{1}{C} \int_0^t i(\tau) d\tau + v_C(0^+)$$

so we evaluate

$$v_C(t) = \frac{1}{C} \int_0^t \frac{5}{3} e^{-8000\tau} \sin(6000\tau) d\tau + 50$$

$$= \frac{5}{3C} \int_0^t e^{-8000\tau} \sin(6000\tau) d\tau + 50$$

recall

$$\int e^{at} \sin(bt) dt = \frac{e^{at}}{a^2 + b^2} [a \sin(bt) - b \cos(bt)]$$

so,

$$v_C(t) = \frac{5}{3C} \left[\frac{e^{-8000\tau}}{(-8000)^2 + (6000)^2} (-8000 \sin(6000\tau) - 6000 \cos(6000\tau)) \right]_0^t + 50$$

$$= \frac{5}{3(10^8)C} [e^{-8000t} (-8000 \sin(6000t) - 6000 \cos(6000t)) + 6000] + 50$$

$$= 50 + 50 + e^{-8000t} \left[-\frac{200}{3} \sin(6000t) - 50 \cos(6000t) \right]$$

thus,

$$\boxed{v_C(t) = 100 - e^{-8000t} \left[\frac{200}{3} \sin(6000t) + 50 \cos(6000t) \right] \text{ V for } t \geq 0}$$

Step 2. Simulate the circuit to confirm the result in Step 1.

[View simulation code](#)

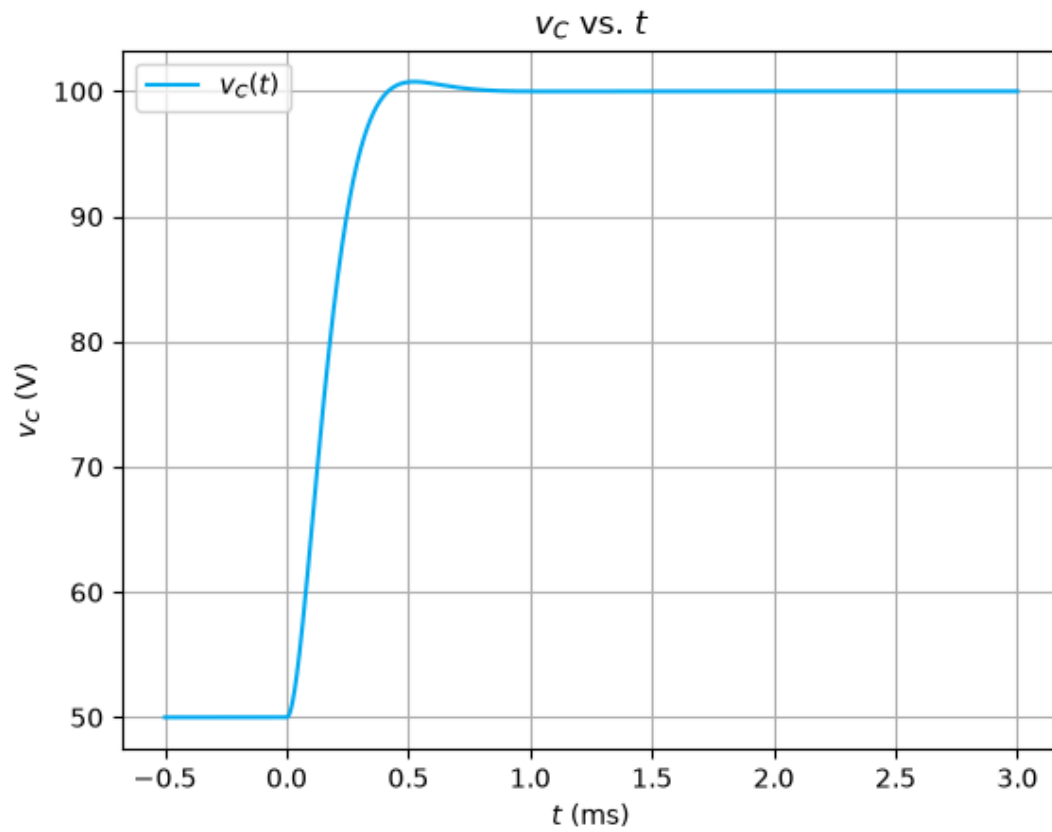


Figure 3: Simulated capacitor voltage $v_C(t)$, confirming the underdamped transient predicted from the analytical solution. The voltage begins at 50 V, briefly overshoots the final value due to the exchange of energy between the capacitor and inductor, and settles to 100 V.